First convergence result

Theorem [Jacot et. al '18, Du et. al '19,

With the initialisation $\begin{cases} w_i \sim \mathcal{N}(0, I_d) \\ a_i \sim \mathcal{N}(0,1) \end{cases}$ for $i = 1, \ldots, m$,

if $m \gtrsim \mathrm{poly}(n)$, then with high probability:

$$L(\theta^t) = \mathcal{O}(e^{-\nu t})$$

✔ training loss converges to 0

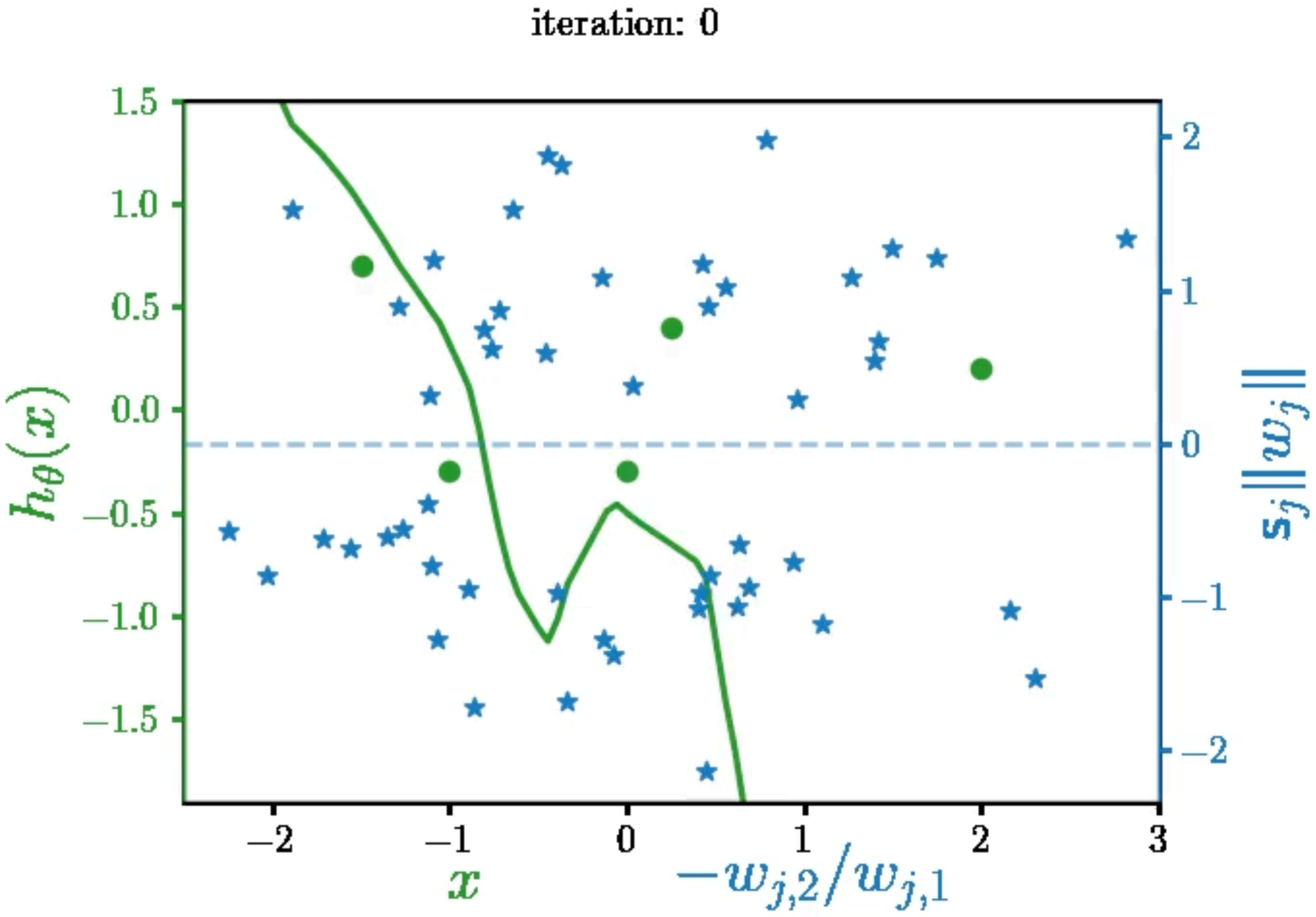
✔ exponential speed of convergence

✘ no feature learning (lazy regime) → due to large initialisation

Illustration of lazy training

Data $x_i \in \mathbb{R}^2$
(bias = second coordinate)

- Dot: data points
- Line: estimated function
- Star: single neuron (w_j, a_j)

  x value $=$ kink position

  y value $= a_j$